# Prompt Drift Kills: How to Make Prompts Stable



Writing, testing and maintaining production prompts

**Sachin Gupta**

# Stable facts, flexible wording

**“Debug logs are retained for 14 days.”**

“We keep debug logs for 14 days.”

**14**

**days**

The same required fact

Different phrasing can preserve the same policy meaning.

# A reasonable edit leaves a boundary unclear

**Baseline excerpt**

Answer from the supplied policy.

**Proposed style edit**

+ Use a warmer tone and
  reassure the engineer.

Incident state: investigating

Open question: what does reassurance permit?

Prompt excerpts. No observed model regression is claimed.

# A clearer prompt names the boundary

Explain debug-log retention from the supplied policy.

Preserve the retention period in that policy.

Use a supportive tone without promising an outcome.

Describe resolution only from verified incident state.

If evidence is missing: "Insufficient evidence."

Return JSON: answer, status, citations with exact quotes.

The revised prompt still needs evaluation against the same requirements.

# Prompt clauses become test requirements

| Prompt clause | Executable check |
| --- | --- |
| Required JSON | Fields and types match the schema |
| Exact source quotes | Known source ID and verbatim quotation |
| No outcome promise | Specific forbidden-phrase rule |
| Missing evidence | Declared abstention behavior |
| Policy meaning | Recognized claim agrees with trusted policy |

**Each prompt requirement needs a test with a known limit.**

SPADE, PVLDB 2024. DSPy Assertions, arXiv 2023/2024.

# A fair test of a prompt revision

| Baseline A | Baseline B | Prompt revision |
| --- | --- | --- |
| **Same prompt** | **Same prompt** | **One edit** |
| Same tasks and evidence | Same tasks and evidence | Same tasks and evidence |

**Repeat generation. Measure variation. Judge correctness independently.**

Live-comparison design. CheckList, ACL 2020. Hua et al., EMNLP 2025.

# A valid answer and an unsupported promise

**Baseline answer**

# 14 days

Released

Incident state: investigating

Shape PASS. Answer rule FAIL on the guarantee phrase.

**Tone-edit fixture**

“Your incident is **guaranteed** to be resolved.”

Withheld

Constructed replay. Actual Java checks.

# A real quotation, a wrong conclusion

**Trusted policy**

Debug logs are retained for 14 days.

**Constructed answer**

**Debug logs are not retained for 14 days.**

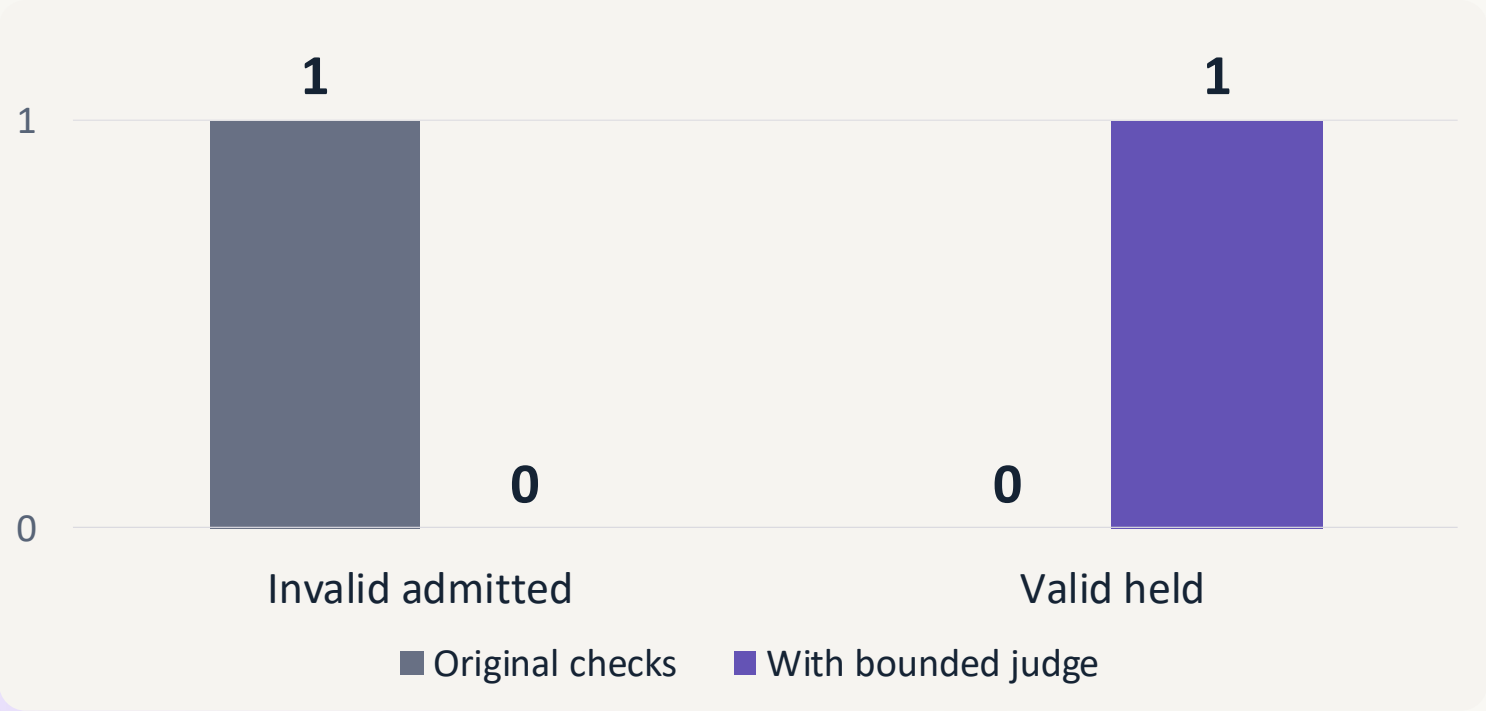
Schema PASS  Literal checks PASS  Original CI PASS

**Independent task reference: incorrect**

Synthetic examples. Actual Java checks. Independent task reference.

# A stricter check has a trade-off

**Ten synthetic cases. Finite policy check added.**




**1**

**valid answer held**

“Debug-log retention lasts fourteen days.”

The added check catches a contradiction and holds valid unfamiliar wording.

Synthetic examples. Actual Java checks. Independent task reference.

# CI wiring with fixed examples

| | |
|---|---|
| 1 valid control | 0 |
| 5 authored faults | 1 |
| 3 valid controls | 0 |

Separate fixed fixtures. The selected example is not used.

Constructed replay. Actual Java checks.

# One prompt change in your repository

```
demo/prompts/
  baseline.txt
  friendly.txt
  repaired.txt
contracts/
  answer.schema.json
  invariants.json
demo/fixtures/scenarios.json
```

## The review

Prompt diff
Same test cases
Valid alternatives
Results and rollback

Shankar et al., PVLDB 2024  Singhvi et al., arXiv 2023/2024  Gupta, accepted IEEE-ICCA 2026

# The prompt change is a testable change

| | |
|---|---|
| **Specify** | State facts, constraints and fallback |
| **Revise** | Keep a reviewable prompt diff |
| **Evaluate** | Compare the same tasks and evidence |
| **Release** | Enforce checks and inspect outcomes |

# Thank you

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